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Repository / Verification Link:

github.com/LAN-SHLOK/SIGNALSCOPE

UNSEEN GENERATOR AUC

0.9170

+19.30% vs Baseline (0.7240)

OVERALL HELD-OUT AUC

0.9986

Cross-Domain Verified

MACRO-F1 SCORE

0.9818

Threshold = 0.50 Calibrated

FALSE POSITIVE RATE (FPR)

1.64%

Accuracy = 98.21% (Balanced)

Model Approach & Dual-Stream Pipeline7,575 Features

Stream 1 (Spatial Foundation):

Frozen DINOv2-reg (ViT-L/14) extracting multi-layer CLS tokens (blocks 8, 16, 20, 24; 4,096-dim) and spatial patch variance matrices (3,073-dim).

Stream 2 (Spectral & Sensor Noise):

Multi-colorspace SRM high-pass residual filters (9-channel), orthonormal 2D FFT radial energy power spectra (128-dim), and Bayer CFA autocorrelation.

Ensemble & Meta-Learner:

Deep multi-task MLP fused with LightGBM gradient boosted trees via a regularized logistic regression meta-learner.

Calibration:

Post-hoc Temperature Scaling (\$T=1.4444\$) yielding an Expected Calibration Error (ECE) of 0.0185 to prevent overconfident accusations.

Dataset & Split ProtocolZero Leakage

Core Benchmark:

CIFAKE dataset (~100k images; CIFAR-10 real photos + Stable Diffusion 1.4 synthetics).

Disclosed Public Additions:

Defactify Image Dataset (MS-COCO real + SD3 Flow-Matching, SDXL, DALL-E 3, Midjourney v6) and uncompressed smartphone crops (Hemg).

Split Ratio:

70% Train (3,382) / 15% Validation (725) / 15% Held-out Test (725). Experience replay buffer employed to eliminate catastrophic forgetting.

Implemented Challenge ModulesCore + 5 Bonuses

Core Task:

Binary Real vs AI Classification with calibrated probabilities.

Module A (Explainability):

60/40 fused DINOv2 attention rollout & SRM Grad-CAM saliency heatmaps with natural language cue descriptors.

Module B (Attribution):

Multi-class generator family identification (Diffusion Latent, Flow Matching, GAN, Real Optical Sensor).

Module C (Robustness):

100% accuracy retention across JPEG re-compression (\$Q=95\$ to \$Q=40\$).

Module D & F (Provenance & UI):

C2PA Manifest parser, deep EXIF camera audit, and full interactive React 19 / Vite workspace.

Quantitative Performance & Baseline DeltaSection 4 Evaluation

Evaluation Target	ViT-B Baseline	SignalScope	Status
Unseen-Split AUC (Primary)	0.7240	0.9170	+19.30%
Overall Held-out AUC	0.9500	0.9986	+4.86%
Macro-F1 Score	0.9000	0.9818	+8.18%
JPEG Degradation (\$Q=40\$)	61.20%	100.0%	Robust
12MP Smartphone Camera	False AI (99%)	0.26% AI	Authentic

Held-out Confusion Matrix & Operating PointThreshold = 0.50

True	Predicted		
	Real	AI	
	TN: 713	FP: 12	
	FN: 14	TP: 711	

Sensitivity (TPR):

98.07%

Specificity (TNR):

98.34%

False Alarm Rate (FPR):

1.64%

Expected Calib. Error (ECE):

0.0185

Disclosed System Limitations & Ethics Compliance:

1. Physical Sensor Interference:

Extreme low-light captures featuring high sensor ISO noise combined with severe motion blur can suppress Bayer CFA demosaicing autocorrelation peaks. The model delegates these edge cases to the Uncertain (Human Review) tier (25%–75% band).

2. Responsible AI Boundary:

Operates strictly under likelihood estimation ("Likely AI-generated") and avoids political claim adjudication or biometric facial tracking per challenge rules.